# Trainings

# Select Service Tools

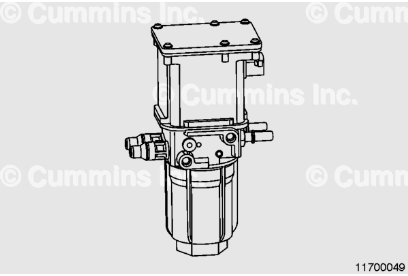
### Recommended Cummins® Service Tool

- Abrasive pad, Part Number 3823258, or equivalent
- Diesel Exhaust Fluid (DEF) Clean Care Kit, Part Number 5394236 & 4919222.

### Additional Service Items

- Super Lube® synthetic grease, or equivalent
- Diesel exhaust fluid dosing unit inlet filter kit.

# General Information




The aftertreatment diesel exhaust fluid (DEF) dosing unit draws DEF from the aftertreatment DEF tank, pressurizes the DEF, and delivers the DEF to the aftertreatment DEF dosing valve. Any unused DEF is then routed back to the aftertreatment DEF tank. See Exhaust System - Overview in Section F for further information on the aftertreatment DEF dosing system.

The DEF dosing unit has two engine coolant connections to allow engine coolant to flow through the DEF dosing unit housing. This allows the engine coolant to thaw the DEF dosing unit and to prevent it from freezing during operation. The coolant supply and return connections are interchangeable. They connect the aftertreatment DEF dosing unit to the engine cooling system.

Use the following procedure for information on handling incorrect or contaminated DEF. Refer to Procedure 011-056 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-056.html)

The aftertreatment DEF dosing unit has a serviceable filter. Refer to Procedure 011-060 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-060.html)

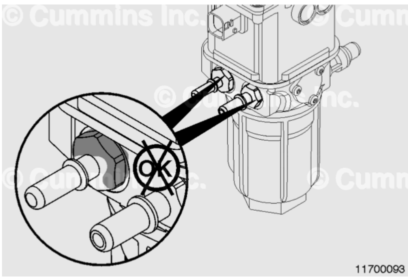
## Initial Check

There is always a chance of residual pressure being present. Open fittings slowly to allow any pressure to bleed off before removing any connections.

Check the DEF lines to and from the dosing unit for signs of leakage. DEF forms a white deposit around leaky fittings.

If white deposits are found, inspect the DEF line connection fittings for damage. See equipment manufacturer service information.

Check around the aftertreatment DEF dosing unit filter cap for white deposits.




## Preparatory Steps

**⚠ WARNING ⚠**

**Diesel exhaust fluid (DEF) contains urea. Do not get the substance in your eyes. In case of contact, immediately flush eyes with large amounts of water for a minimum of 15 minutes. Do not swallow. In the event the DEF is ingested, contact a physician immediately. Reference the Materials Safety Data Sheet (MSDS) for additional information.**

**⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

**⚠ CAUTION ⚠**

**Do not pressure wash or steam clean this unit. Pressure washing or steam can damage the unit. Use compressed air to remove any loose debris.**

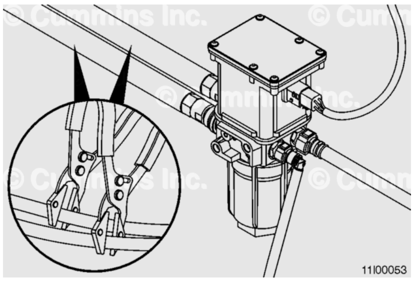
Allow for DEF to recirculate and depressurize after engine shut down and prior to DEF system servicing. This can take up to 15 minutes.

There is **always** a chance of residual pressure being present. Open fittings slowly to allow any pressure to bleed off before removing any connections.

- Disconnect the batteries. See equipment manufacturer service information.

## Remove

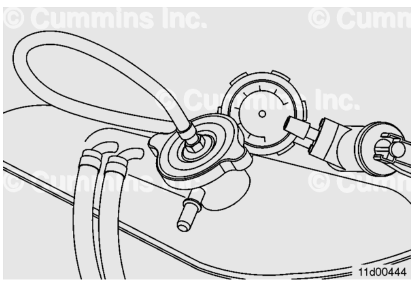
Do **not** drain the cooling system. Use Method 1 or Method 2 to prevent draining the cooling system when the aftertreatment DEF dosing unit coolant lines are disconnected.




### Method 1

- **Note : Prior to pursuing this method, inspect the aftertreatment DEF dosing unit coolant lines to make sure they are flexible hose material that will not be damaged when using the coolant hose pinch-off pliers.**
- Use a pair of coolant hose pinch-off pliers. Clamp both of the aftertreatment DEF dosing unit coolant lines.
- Place a container under the aftertreatment DEF dosing unit. A small quantity of coolant may drain from the coolant lines when disconnected.
- **Note : The coolant supply/return and DEF lines are supplied by the original equipment manufacturer (OEM). The removal of these lines may vary by OEM. The fittings on the aftertreatment DEF dosing unit are set up for quick disconnect style fittings.**
- Disconnect the coolant lines. See equipment manufacturer service information.

### Method 2

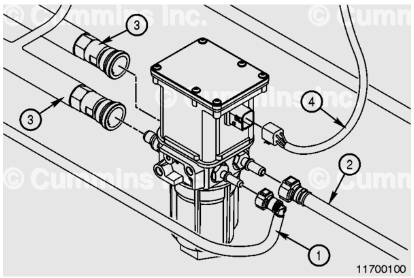



- Determine the appropriate radiator/coolant expansion tank cap adapter for the vehicle being serviced. The list of available caps and applications can be found in Section 14 of the Service Products Catalog.
- Install the adapter on the radiator/coolant expansion tank. Connect an automotive hand-held vacuum pump and

apply 76 mm Hg [3 in Hg] of vacuum.

- **Note : Depending on the cooling system configuration, more vacuum may be required to keep the cooling system in balance, especially on systems where there is a significant height difference between the aftertreatment DEF dosing unit and the radiator/coolant expansion tank fill location**
- **Note : The coolant supply and DEF lines are supplied by the OEM. The removal of these lines may vary by OEM. The fittings on the aftertreatment DEF dosing unit are set up for quick disconnect style fittings.**
- Place a container under the aftertreatment DEF dosing unit. A small quantity of coolant may drain from the coolant lines when disconnected.
- Disconnect the coolant lines. See equipment manufacturer service information.

**Note : The DEF lines are connected to the dosing unit by quick release fittings that could vary between vehicle manufacturers. See equipment manufacturer service information.**

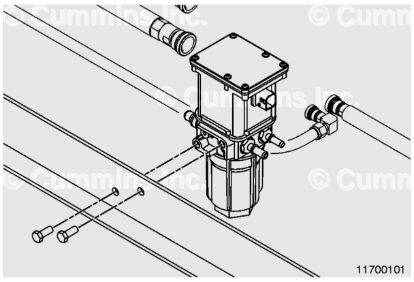



Disconnect the DEF lines and electrical connector in the following order to reduce the possibility of accidental DEF contamination of the electrical connector:

1. Inlet: DEF supply to the aftertreatment DEF dosing unit
2. Outlet: DEF supply to the aftertreatment DEF dosing valve
3. Coolant connectors
4. 4 pin electrical connector.

Cap the DEF and electrical connectors of the aftertreatment DEF dosing unit to prevent contamination and DEF leakage.

Remove the two mounting capscrews.

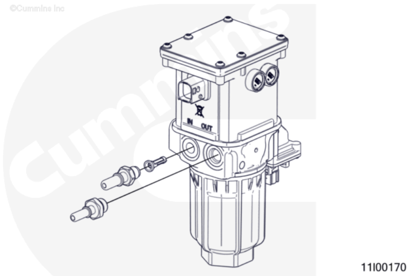



Remove the aftertreatment DEF dosing unit.

# Disassemble

⚠ **CAUTION** ⚠

**Do not screw the M5 screw into the dosing unit housing. Only thread the screw into the screen filter one to two turns.**



**Note : The aftertreatment DEF dosing unit must only be disassembled if a symptom has been identified that indicates further investigation is required.**

**Note : Use the following procedure if the aftertreatment DEF dosing unit filter needs to be removed. Refer to Procedure 011-060 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-060.html)**

Remove the DEF connector(s). Verify the o-ring(s) are attached to the connector(s) and are **not** stuck inside the DEF dosing unit.

Remove the inlet DEF connector and inspect for debris. If debris is found, remove the inlet connector screen filter.

Use an M5 screw to thread into the screen filter. **Only** thread the screw into the screen filter one to two turns. Pull to remove the inlet connector screen filter from the pump housing.

Discard the inlet screen.

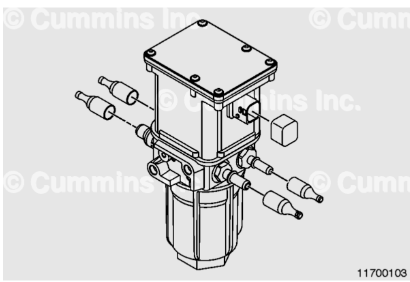
Remove the outlet DEF connector.

Remove the screw of the coolant connector.

Remove the coolant connectors. The coolant connector is a one-piece assembly.

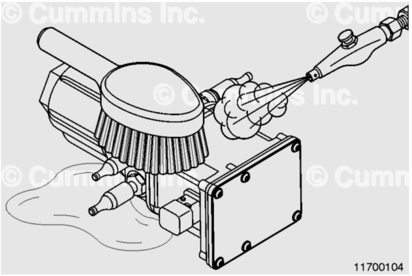
# Clean and Inspect for Reuse

Install protective caps on the fluid connector(s) and the electrical connector on the aftertreatment DEF dosing unit. Use the DEF Clean Care Kit, Part Numbers 5394236 and 4919222.



⚠ WARNING ⚠

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

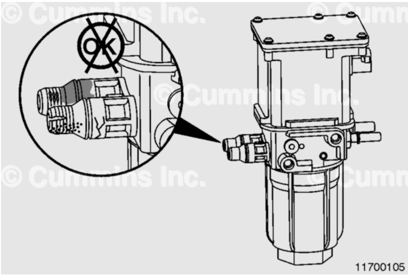



**Note : This unit is not serviceable. Do not open the case.**

Clean the aftertreatment DEF dosing unit using **only** warm water and a soft bristled brush.

Thoroughly clean the areas surrounding the inlet connector, outlet connector, and main filter cap.

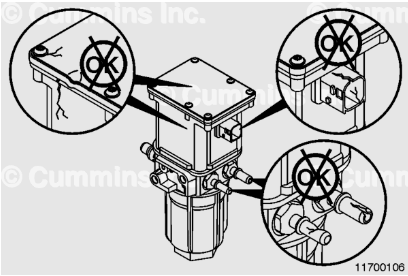
If corrosion is found, the coolant can be contaminated and/or the concentration incorrect. Check the coolant. For specifications, refer to service bulletin Cummins® Coolant Requirements and Maintenance, Bulletin 3666132 (/qs3/pubsys2/xml/en/bulletin/3666132.html).




Clean with an abrasive pad, Part Number 3823258, or equivalent, and a clean cloth.

If the pitting and/or corrosion can **not** be removed with the abrasive pad, replace the coolant connector.

Inspect the outside of the unit. If there are any cracks or other damage to the exterior of the case or electrical connectors, replace the dosing unit.



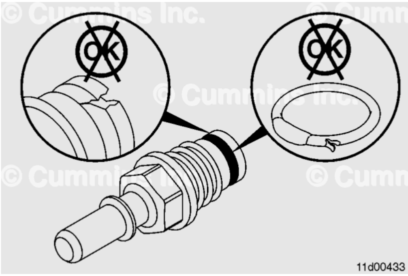

If there are any cracks or other damage to the DEF quick disconnect connectors, replace the DEF quick disconnect connectors.

Use the following procedure if the aftertreatment DEF dosing unit filter cap is damaged. Refer to Procedure 011-060 in Section 11. (/qs3/pubsys2/xml/en/procedures/493/493-011-060.html)

Inspect the DEF line connection ports for cracks or pitting.

If the aftertreatment DEF dosing unit connectors were removed, inspect for cracks and broken o-rings.

Replace the aftertreatment DEF dosing unit connectors or o-rings, if damaged.

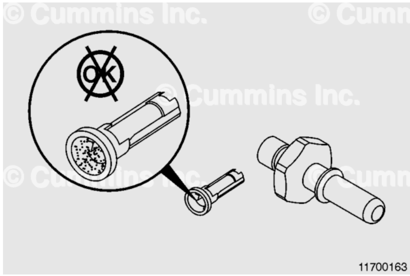



The DEF inlet connector on the aftertreatment DEF dosing unit has an incorporated serviceable filter screen.

Inspect the inlet connector screen.

Install a new screen.

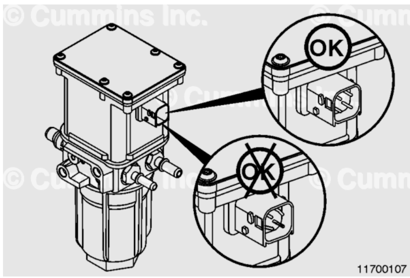
If debris is found in the inlet connector filter screen, inspect the aftertreatment DEF tank filter. See equipment manufacturer service information.




Inspect the electrical connection of the aftertreatment DEF dosing unit for signs of corrosion and debris buildup.

If the pins are heavily corroded or damaged, replace the aftertreatment dosing unit.

If the electrical connector on the aftertreatment DEF dosing unit is damaged, also inspect the 4 pin electrical connector wiring harness. See equipment manufacturer service information.




# Assemble

**Note : The inlet and outlet connectors are not interchangeable and can only be installed one way due to thread and length restraints.**

If necessary, install a new DEF inlet screen filter.

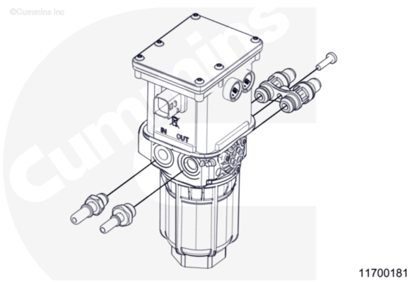
Install a new DEF inlet connector.

**Torque Value:** 14 n•m [ 124 in-lb ]

Install a new DEF outlet connector.

**Torque Value:** 20 n•m [ 177 in-lb ]

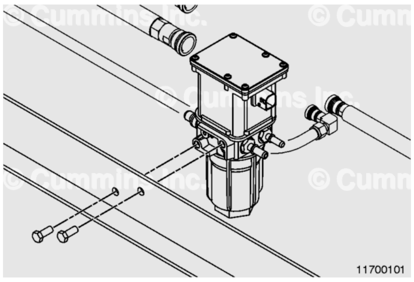
Install the coolant connectors.



Push the coolant connectors into the cavities. Tighten the coolant fittings capscrew.

**Torque Value:** 7 n•m [ 53 in-lb ]

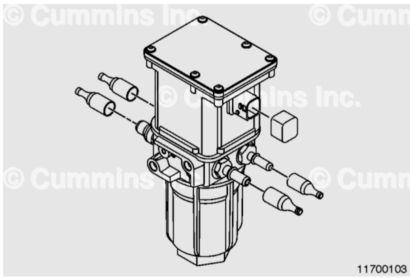
## Install




**Note : Verify the aftertreatment DEF dosing unit remains free from contamination during installation to the vehicle. To reduce the possibility of false DEF leak reports, use a clean, damp cloth to wipe away any DEF that spilled during the repair process.**

Attach the aftertreatment DEF dosing unit to the OEM mounting bracket and secure with the two capscrews.

**Torque Value:** 18 n•m [ 159 in-lb ]




DEF forms a white residue when allowed to dry. This residue can prevent proper sealing of the fittings when the DEF line is re-installed. Before making the DEF line connection:

- Dissolve DEF residue off the DEF line fittings with distilled water.
- Inspect the DEF line fitting o-rings for cuts or damage. Replace the DEF lines if there are any damage to the o-rings.
- Lubricate the DEF line fittings with a drop of synthetic grease such as Super Lube®.
- Remove the caps from the electrical and DEF connectors.
- Install the electrical connector. DEF will cause the electrical connector pins to corrode.
- Install the DEF supply line inlet connector
- Install the DEF pressure line outlet connector
- Install the coolant connectors.

Gently pull on the connectors to confirm they are secured.

## Finishing Steps

**⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

**⚠ CAUTION ⚠**

**Do not use the flow test portion of recommended Cummins® electronic service tool or equivalent DEF Doser Pump Override Test to check the system for leaks. This will spray DEF into the exhaust system at temperatures too low to evaporate, resulting in deposit formations in the exhaust system.**

- Fill the engine with coolant as needed. See Cooling System in Section 8.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

**Last Modified: 17-Mar-2021**